

Figure 9.5 US Perdix micro-drone swarm deployment. ⏎

9.8.6 China – Zhuhai air show swarm demonstrations

China has been a prominent leader in showcasing large-scale drone swarms, with the Zhuhai Air Show serving as the primary venue for public demonstrations of its advanced swarm technology. At these exhibitions, hundreds to over a thousand drones have been launched simultaneously, performing highly synchronized aerial patterns, formation transitions, and cooperative maneuvering routines. While these displays often appear artistic in nature, they are underpinned by sophisticated mesh communication architectures, distributed AI control systems, and real-time swarm coordination algorithms, reflecting dual-use capabilities relevant to both civilian and military domains.

The Zhuhai demonstrations highlight China’s emphasis on scalable swarm deployment, where the primary focus is not on the capabilities of any individual drone, but on the collective behavior of the group. Each drone operates with localized autonomy, continuously exchanging positional and state data with neighboring units to maintain formation